

RESEARCH PLATFORM OVERVIEW · SEPTEMBER 2026

A governed, multimodal precision-intelligence platform

A technical overview for biomedical researchers and principal investigators.

Working platform · Controlled access · Synthetic demonstration data · No claim of independent clinical validation

GENÉTICA

O genoma que orienta a conduta

FTO · DIO2 · IL6

METABOLÔMICA

O metabolismo em tempo clínico

Glicina · Lactato ·
Acilcarnitinas

Structure before synthesis

BioSync turns molecular and laboratory inputs into structured, provenance-carrying findings, then permits cross-domain synthesis only after each domain has resolved its own canonical truth.

No domain is allowed to repair another domain's uncertainty by inference.

01

Resolve

Normalize source data and determine the domain-specific evidence state.

02

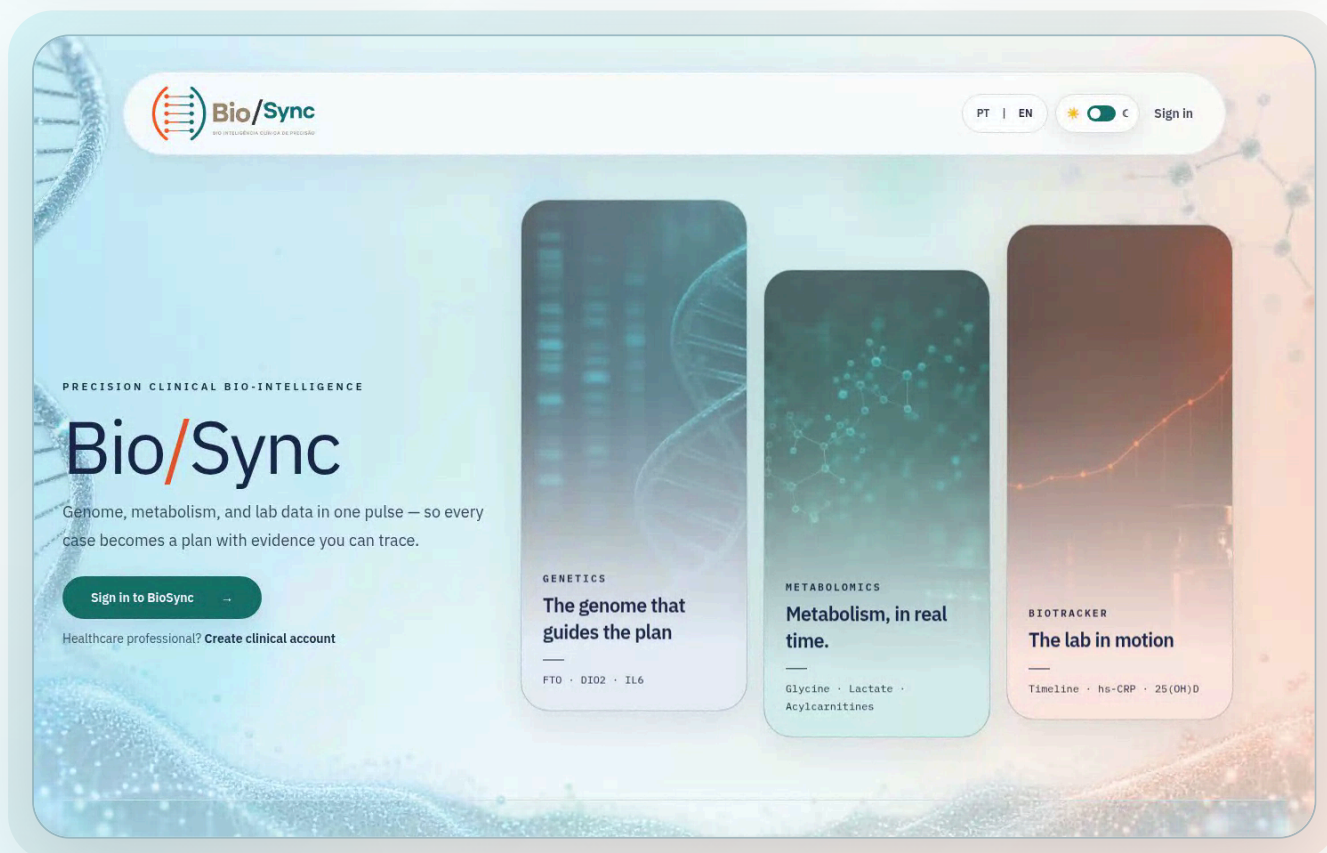
Govern

Keep candidate science separate from signed, canonical content.

03

Converge

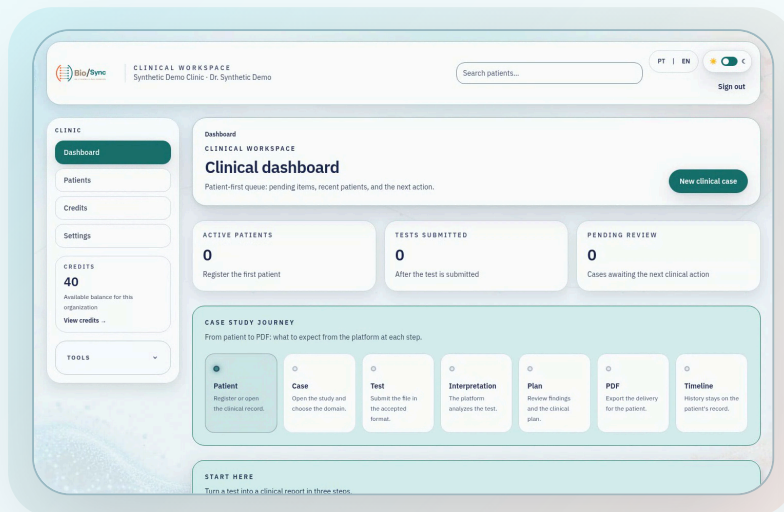
Synthesize only resolved outputs while preserving conflicts and insufficiency.



Independent domains, shared governance



From identity to reviewable output



1. Establish identity

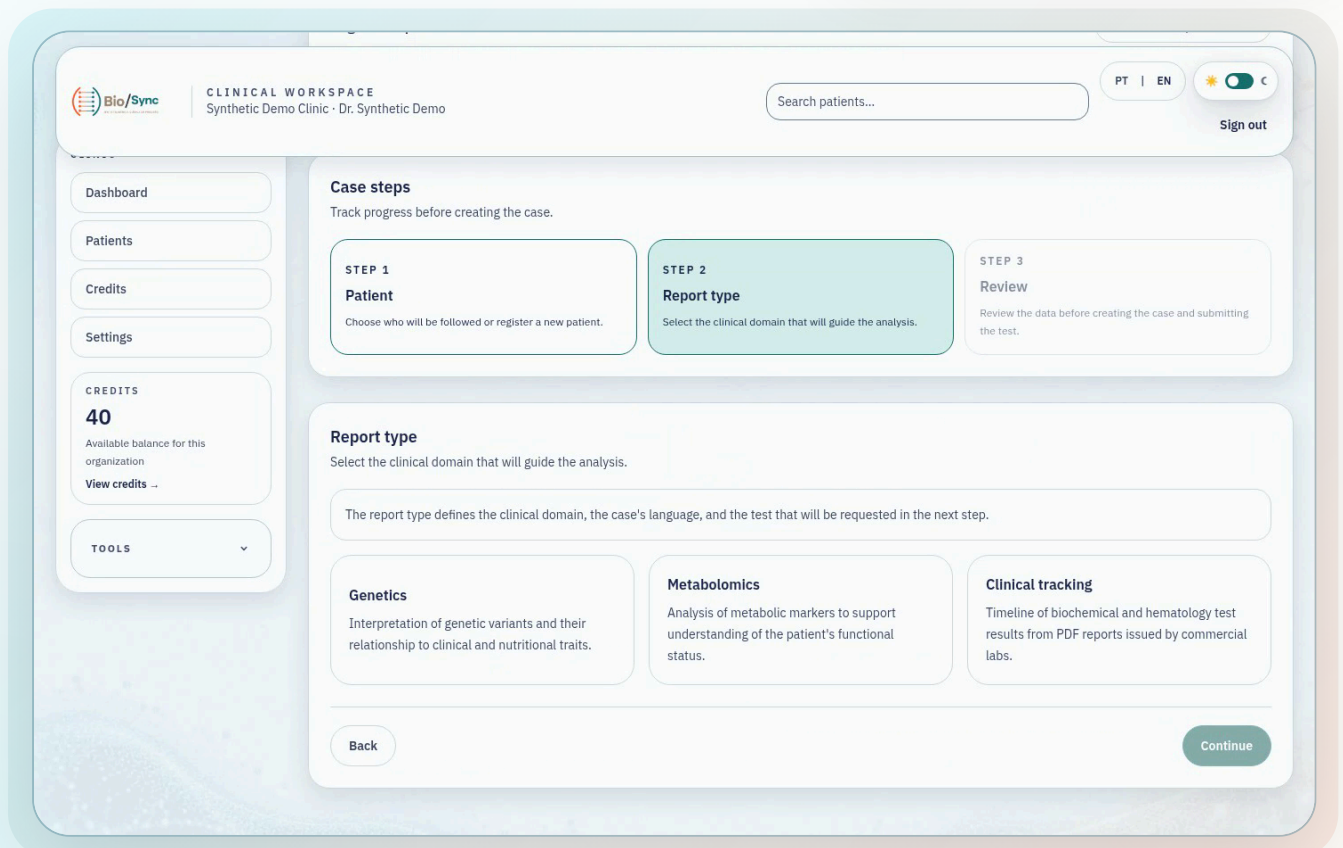
A patient or research subject record anchors every downstream case, test, interpretation, and report.

2. Create the case

The responsible professional and scientific domain are recorded before interpretation begins.

3. Attach evidence

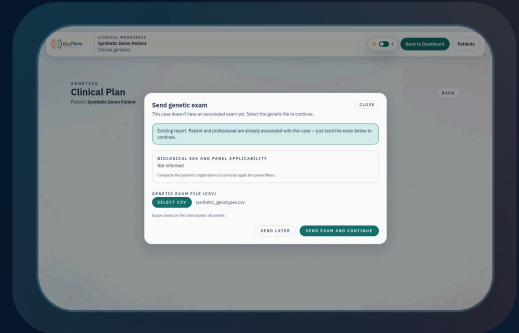
The raw source remains associated with the existing case; processing and export do not create disconnected duplicates.



Raw-supported findings, explicit technical coverage

The genetics pipeline resolves observed calls against a curated authority before any patient-facing narrative is produced.

- Raw genotype presence and exact observed-genotype consistency remain auditable.
- Panel inclusion is kept distinct from raw-data-supported findings.
- Resolved, unverified, and absent components remain separate technical states.
- Unresolved strand orientation and compound models fail closed.
- Guidance may be panel-level without becoming genotype-prescriptive.



**The renderer presents resolved science;
it does not create it.**

Observed

What the source file actually contains.

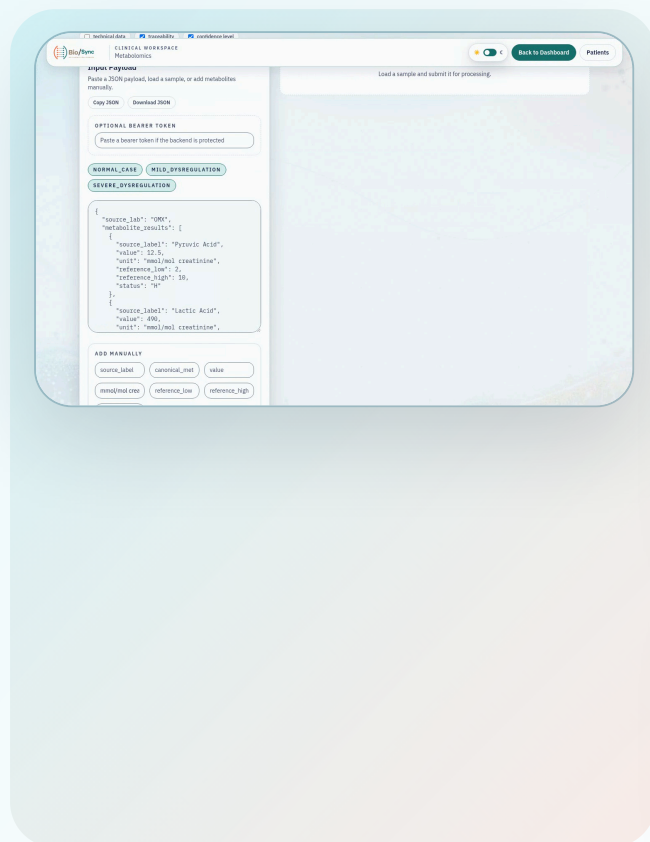
Resolved

What the canonical model can support.

Projected

What may safely become visible to the professional.

Normalize without losing provenance



Canonical analytes

Aliases and source-laboratory labels map into a source-independent model while retaining the original source.

Unit-aware interpretation

Normalization records unit context and conversion confidence instead of treating incompatible values as equivalent.

Explicit safety states

Accepted, flagged, discarded, missing, and insufficient are meaningful outcomes—not rendering errors.

Layer	Responsibility	Preserved evidence
Adapter	Understand the source format	Source label and source document
Canonical registry	Resolve analyte identity	Alias and mapping path
Normalization	Reconcile units and ranges	Original value, normalized value, confidence
Pattern layer	Evaluate domain-specific structures	Inputs, rule, outcome, limitation

Synthesis without evidence substitution

Bio/Sync CLINICAL WORKSPACE
Integrated analysis

Integrated Analysis

Review findings, convergence, and recommendations generated by the clinical fusion engines.

Select an analysis, load a scenario, and run it to review findings and recommendations.

Integrated Analysis

IRON_SUPPORT METHYLATION

INFLAMMATORY WEAK_NOISE

BEARER TOKEN

Optional

CLINICAL MODE **ADVANCED MODE**

CBC

Paste one marker per line. Example: Hemoglobin: 11.3

Hemoglobina: 12.1
Hematocrito: 36
Eritrocitos: 4
VCM: 103.5
RDW: 15

METABOLOMICS

One metabolite per line. Example: Methylmalonic Acid: high or
Methylmalonic Acid: 6.0 | mmol/mol creatinine | ref 0.5-3.5 | high

Methylmalonic Acid: 6 | mmol/mol creatinine | ref 0.5-3.5 | H
FIGLU: 4.2 | mmol/mol creatinine | ref 0.1-2 | H

Allowed

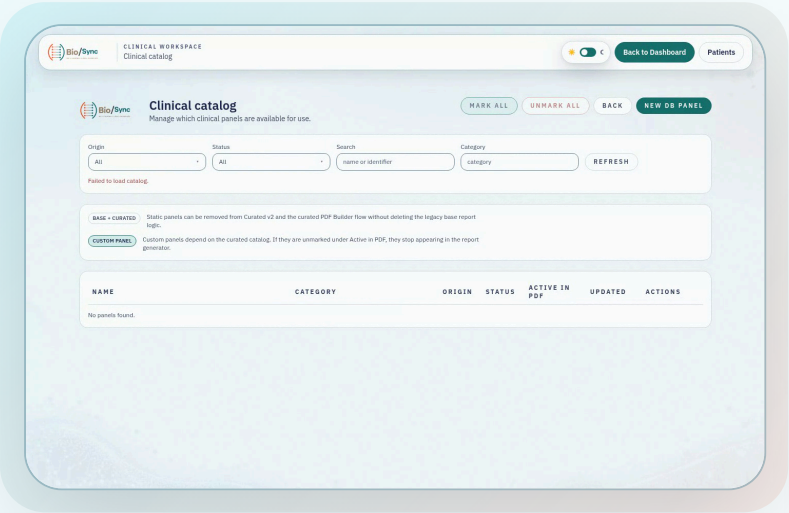
- Identify support across independently resolved domains.
- Expose disagreement and temporal context.
- Return insufficient when the minimum evidence base is absent.

Prohibited

- Infer a missing genotype from a biomarker.
- Rewrite a resolved domain result retrospectively.
- Convert correlation into autonomous diagnosis or treatment.

Convergence is a classified relationship between evidence states—not a license to fill evidentiary gaps.

The human signature is the final gate



Automation may gather evidence, classify candidate associations, and identify ambiguity. It cannot promote a candidate to canonical content.

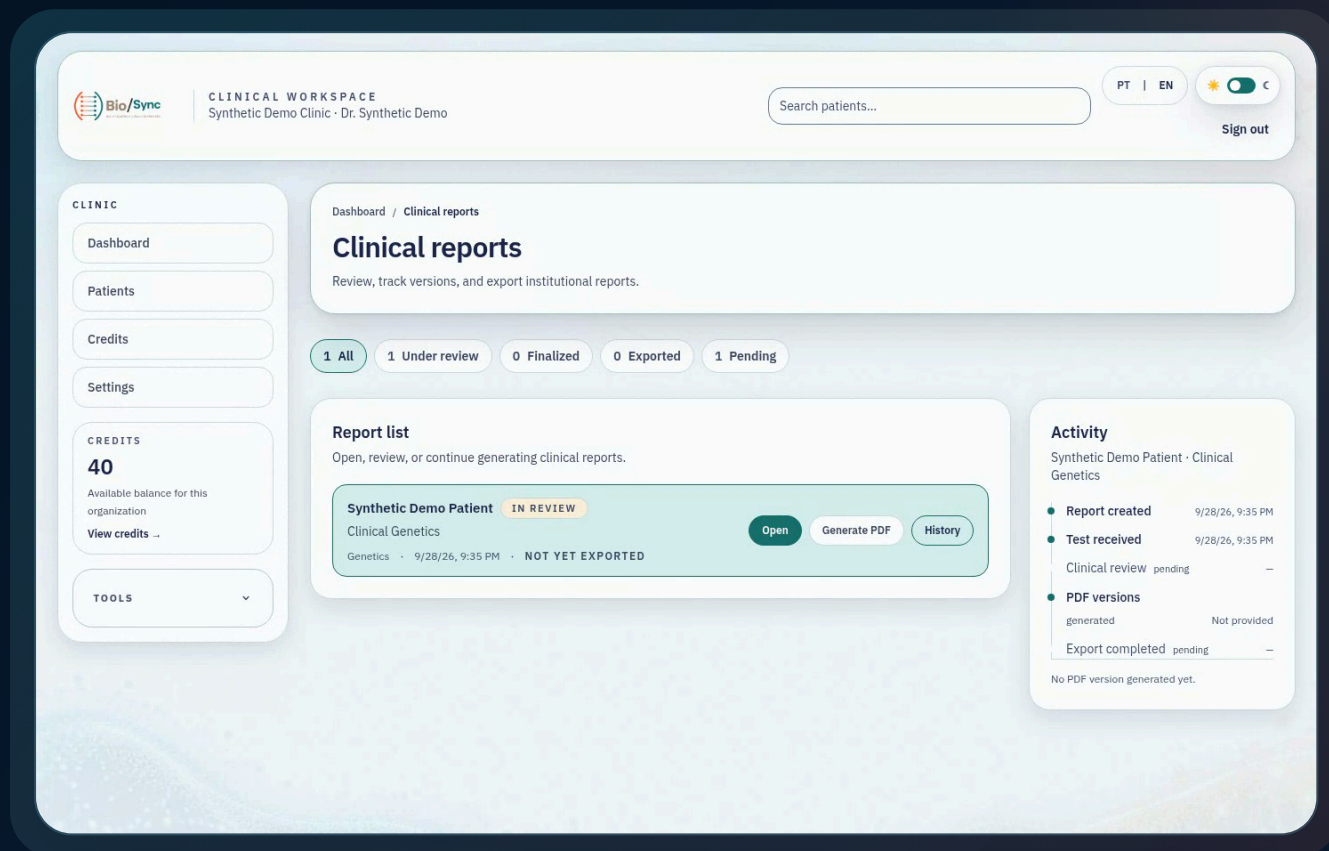
`pending_adjudication`
No human decision. Not canonical science.

`candidate*_pending_human_signature`
Prepared for review. Not approved.

`signed_approved_for_yaml*`
Explicit reviewer, decision, and date recorded.

Activity	Automation	Human authority
Evidence gathering	Assisted	Reviews relevance and sufficiency
Preliminary classification	Assisted	Resolves ambiguity
Final signature	Never	Always explicit
Canonical promotion	Never automatic	Recorded decision required

Auditable output, governed presentation



Versioned

Each export remains attributable to its case and generation event.

Canonical

Real and Gold outputs use the same renderer and visual components.

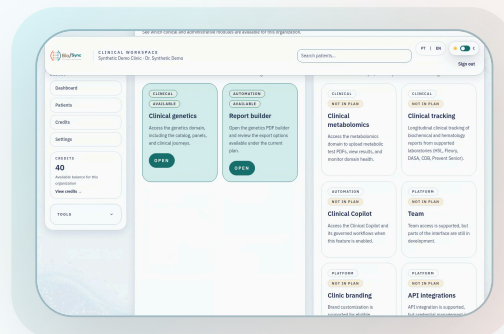
Separated

Projection decides content; rendering decides presentation.

Deterministic synthetic fixtures support regression review without exposing real patient artifacts. Visual baselines govern presentation only; they never authorize or alter scientific content.

Controlled by design

- Real patient data does not enter version control, fixtures, screenshots, or visual baselines.
- Demonstration material uses fictional identities and synthetic molecular inputs.
- Capabilities are explicit by tenant and role; access is not inferred from navigation.
- Clinical storage is separated from application artifacts and protected by operational backup and integrity checks.
- Third-party AI access is subject to provider-specific privacy policy and human-gated workflow controls.



A missing permission, missing file, or ambiguous result is a state to expose—not a reason to guess.

0

PHI in this package

0

Production writes during capture

100%

Synthetic demonstration identities

A multi-omic platform designed to keep advancing

Scientific authorities

Canonical registries, evidence thresholds, and candidate-to-signed promotion criteria evolve under explicit curation and review.

Domain engines

Genetics, metabolomics, laboratory, and convergence capabilities advance independently. Microbiota is the next domain in development.

Validation frameworks

Evaluation protocols can mature with defined endpoints, inclusion criteria, comparators, and uncertainty reporting.

Human factors

Traceability, review burden, comprehension, and professional oversight remain first-class design concerns.

Reproducibility

Synthetic fixtures, deterministic engines, and report-generation baselines are updated alongside the product.

Governance

Human gates, auditability, tenant controls, privacy boundaries, and data handling evolve with operational requirements.

Continuous evolution must strengthen traceability and human review—not bypass them.

DECLARED LIMITATIONS

What is not being claimed

Public availability

BioSync is not yet publicly available to health professionals. There is no open trial, public demonstration tenant, or announced public release date.

Independent validation

This document does not claim independent clinical validation, diagnostic accuracy, comparative effectiveness, or regulatory clearance.

Uniform domain maturity

Domain maturity differs. Genetics is the most mature; other engines continue to evolve, with microbiota now in development as the next domain.

Autonomous practice

Outputs are decision-support artifacts for professional review. They are not autonomous diagnoses or prescriptions.

DEVELOPMENT POSTURE

Evolving under explicit scientific and operational controls

New evidence, methods, domain capabilities, and interface refinements are incorporated through governed, traceable change. Microbiota is the next domain currently in development.

MICROBIOTA NEXT

HUMAN REVIEW

VERSIONED CHANGE

REPRODUCIBILITY

Prepared for academic discussion. This overview describes the platform's current architecture and declared boundaries; capabilities and maturity continue to evolve.